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Adaptive Human-AI Shared Autonomy Frameworks for Dynamic Multi-Agent Decision Environments

Elena Rostova • Third-Year PhD Candidate

Primary Advisors: Prof. Marcus Vance & Dr. Sophia Al-Mansoor

 Institute of Advanced Technology  elena.rostova@iat-edu.org  elena-rostova.research.org

Executive Summary & Research Abstract

High-stakes multi-agent environments—such as autonomous fleet management, robotic disaster response, and dynamic network defense—increasingly rely on hybrid human-AI teams. However, conventional shared autonomy frameworks struggle with two critical flaws: static allocations of authority that fail under non-stationary task demands, and cognitive overload induced when human operators must intervene during unexpected agent policy drifts. This dissertation proposal presents **SYNAPSE-LOOP**, a novel adaptive shared control paradigm that dynamically models human cognitive state, predicts intervention intent, and negotiates task authority in real time. By fusing bi-directional intent forecasting with multi-agent reinforcement learning (MARL), our framework maintains mutual alignment while minimizing unnecessary operator burden. Early empirical evaluations on simulated emergency logistics demonstrate a 34% reduction in human cognitive workload metrics alongside a 28% improvement in mission completion velocity compared to standard supervisory baselines.

Candidacy Snapshot

Institution: Inst. of Advanced Tech.**Department:** Computer Science & AI**PhD Start Date:** September 2023**Expected Graduation:** May 2027**Candidacy Status:** Proposal Approved**Primary Area:** Human-AI Collaboration, MARL, Shared Autonomy**Research Keywords:**

- Shared Autonomy
- Human-in-the-Loop
- MARL Policy Shift
- Cognitive Workload

1 Problem Statement & Research Motivation

Autonomous multi-agent systems are rapidly transitioning from controlled laboratory settings into dynamic, high-uncertainty operational environments. In scenarios such as robotic search-and-rescue, urban aerial mobility management, and dynamic supply chain dispatch, full agent autonomy remains unfeasible due to unpredictable environmental edge cases and complex ethical trade-offs. Consequently, system architects implement *Human-in-the-Loop* (HITL) architectures to retain human judgment for high-level strategic supervision.

Despite significant advances in single-agent shared control, existing multi-agent supervisory frameworks suffer from three systemic limitations:

- P1. Rigid Authority Boundaries:** Existing frameworks rely on pre-scripted state transitions (e.g., manual override toggles), which lack the granularity required to transfer partial authority across heterogeneous sub-tasks under fluctuating network latency.
- P2. Cognitive Context Disconnect:** When autonomous agents encounter out-of-distribution states and request human intervention, operators suffer severe situational awareness degradation (*out-of-the-loop latency*), frequently leading to suboptimal or contradictory control inputs.